

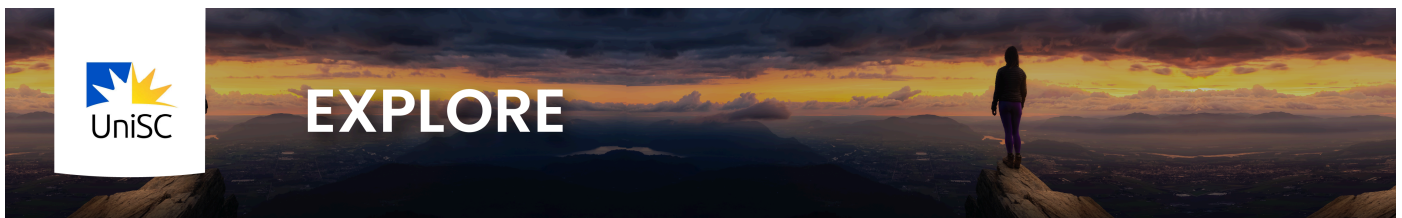
3.5 Downside Risk



Downside Risk

Guiding Questions

- Why might standard deviation underestimate investment risk?
- Why do investors care more about downside risk than upside volatility?
- How does the Sortino ratio improve on the Sharpe ratio?



What will you learn?

You will learn:

- Why traditional volatility measures may not fully capture investment risk.
- How Sortino Ratio focuses specifically on harmful volatility.
- Why downside risk measures are important for risk management and portfolio construction.

1. Why Do Investors Care About Downside Risk?

In previous sections we measured risk using standard deviation, which captures the overall variability of investment returns. However, standard deviation treats positive and negative fluctuations equally.

For example:

- a return of +15%
- a return of -15%

Both contribute equally to standard deviation because they represent deviations from the average return.

In reality, investors usually do not consider positive deviations to be harmful. Investors are primarily concerned about losses or negative outcomes.

For this reason, financial analysts often examine downside risk, which focuses specifically on the possibility of negative returns.

Understanding downside risk helps investors:

- evaluate potential losses more accurately
- assess the likelihood of poor investment outcomes
- design portfolios that limit severe losses
- better understand extreme market events

Downside risk analysis therefore complements traditional risk measures such as standard deviation.



Watch

[Sortino Ratio: Introduction](#)



Details

Transcript



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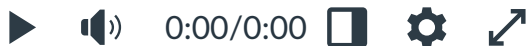
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Estimating the Sortino Ratio in R



Details

Transcript



[R Script \(https://learn.usc.edu.au/courses/35359/files/3294288?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3294288?wrap=1) 
(https://learn.usc.edu.au/courses/35359/files/3294288/download?download_frd=1)

2. Downside Deviation

A common way to measure downside risk is downside deviation, which considers only returns that fall below a specified target or minimum acceptable return.

Instead of measuring variability around the mean return, downside deviation measures variability below a target return level.

The target return is often referred to as the Minimum Acceptable Return (MAR).

This approach focuses only on outcomes that are considered undesirable by investors.

Downside Deviation Formula

$$\text{Downside Deviation} = \sqrt{\frac{\sum (\min(0, R_i - MAR))^2}{n}}$$

Where

R_i = individual return

MAR = minimum acceptable return

n = number of observations

Returns above the target level do not contribute to the downside deviation calculation.

3. The Sortino Ratio

The Sortino ratio is a performance measure that evaluates return relative to downside risk, rather than total risk.

It is similar to the Sharpe ratio introduced earlier, but it replaces standard deviation with downside deviation.

Sortino Ratio Formula

$$\text{Sortino Ratio} = \frac{R_p - \text{MAR}}{\text{Downside Deviation}}$$

Where:

R_p = portfolio return

MAR = minimum acceptable return

The Sortino ratio therefore measures the excess return earned for each unit of downside risk.

Example: Calculating the Sortino Ratio

Suppose a portfolio has the following characteristics:

Average return = 12%

Minimum acceptable return (MAR) = 5%

Downside deviation = 6%

Step 1: Calculate excess return

$$12\% - 5\% = 7\%$$

Step 2: Divide by downside deviation

$$\text{Sortino} = 7/6 = 1.167$$

Interpretation

A Sortino ratio of 1.17 indicates that the portfolio generates 1.17 units of excess return for each unit of downside risk.

Higher Sortino ratios indicate better performance relative to downside risk.

Because the Sortino ratio focuses only on harmful volatility, many investors consider it a more realistic measure of investment risk.

4. Importance of Tail Risk in Financial Markets

Financial markets occasionally experience extreme events such as:

- financial crises
- sudden market crashes
- systemic banking failures
- geopolitical shocks

These events occur more frequently than predicted by standard statistical models.

Because standard deviation and even VaR may underestimate these extreme outcomes, risk managers often analyse tail risk to understand potential catastrophic losses.

Understanding tail risk helps investors:

- design more resilient portfolios
 - prepare for extreme market conditions
 - manage risk exposure more effectively
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5. Key Insights from Downside Risk Analysis

From this section we learn:

- Standard deviation treats positive and negative volatility equally
- Investors are usually more concerned about negative outcomes
- Downside deviation measures risk below a minimum acceptable return
- The Sortino ratio evaluates return relative to downside risk

Downside risk measures therefore provide additional insight beyond traditional risk measures.

Reflect

Why might standard deviation fail to capture the risk that investors care most about?

What is the difference between downside deviation and standard deviation?

What does the Sortino ratio measure?

Solve

Question 1

A portfolio has:

Average return = 14%

Minimum acceptable return = 6%

Downside deviation = 8%

Calculate the Sortino ratio.

Question 2

A portfolio recorded the following six quarterly returns:

Q1: 7%, Q2: -4%, Q3: 11%, Q4: 2%, Q5: -8%, Q6: 14%

The minimum acceptable return (MAR) is 3%.

1. Calculate the downside deviation.
 2. Calculate the average return and the excess return.
 3. Calculate the Sortino ratio.
 4. Interpret what the result means for this portfolio.
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Answers available [here \(https://learn.usc.edu.au/courses/35359/files/3292656?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3292656?wrap=1).

Critically evaluate

Suppose an investment portfolio has a low standard deviation but occasionally experiences very large losses.

Discuss why relying only on standard deviation might underestimate the true risk of this portfolio.